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Robustness-Driven Feature Attribution in Genomic Deep Learning with Integrated Gradients

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Abstract

Deep learning models can predict phenotypes from genomic sequences, but gradient-based attributions are often noisy and only partially aligned with biologically meaningful features. In computer vision, adversarial training has been shown to improve attribution quality under specific structural assumptions. Whether these benefits transfer to discrete genomic data, and under what conditions, remains unclear.

This thesis investigates when adversarial robustness improves interpretability for sequence-based convolutional neural networks in bacterial genomics, focusing on Integrated Gradients as the attribution method. We propose a theoretical framework specifying when adversarial training can improve attribution quality: it requires valid ground truth (evaluability), learnable causal features that are more robust than confounders (identifiability), and gradients that faithfully reflect decision logic (faithfulness). Adversarial training primarily improves identifiability by suppressing non-robust confounders; boundary-aware attribution improves faithfulness independently.

We test the framework through three tasks. On synthetic data where compositional shortcuts, e.g., GC content, compete with planted motifs, adversarial training recovers the true signal when shortcuts are present but not overwhelming. On an external benchmark without dominant compositional confounders, a robust attribution aligns saliency with the decision boundary normal and hence provides the primary interpretability gain, while a robust model through adversarial training contributes secondary noise reduction. On real bacterial genomes, lightweight diagnostics detect strong compositional shortcuts and predictive multiplicity, explaining near-chance attribution despite high predictive accuracy.

We propose a conceptual factorization that organizes these findings, showing how evaluability, identifiability, and faithfulness jointly gate attribution quality. A key additional finding is that explanation multiplicity poses a fundamental challenge for genomic model interpretability.